

Intro to Neural Networks

2026.04.30

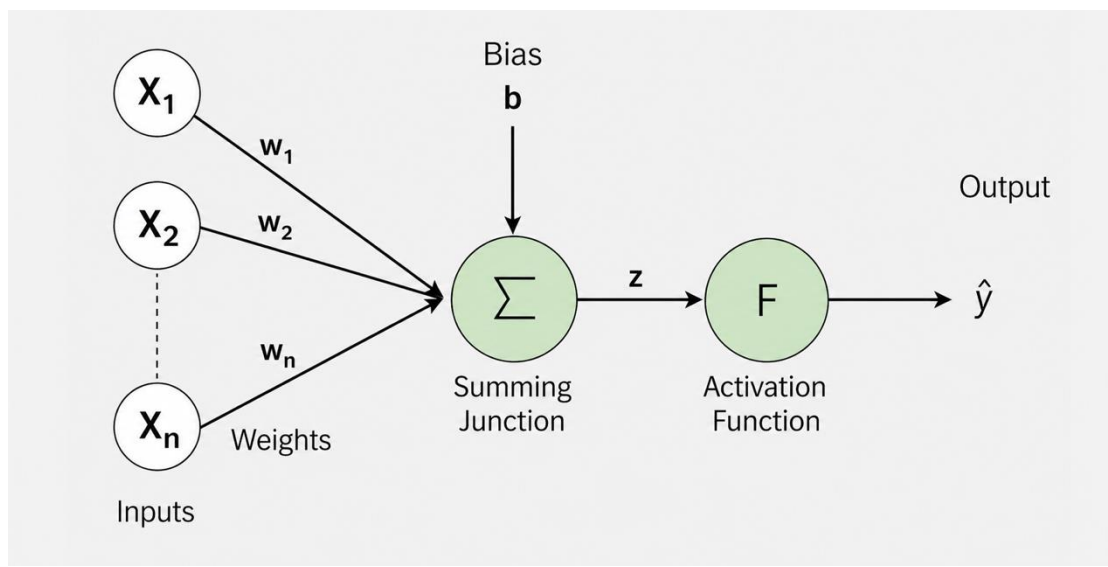
Midterm next week

- No in person class
- Conceptual portion
 - Available Tuesday noon-8 pm in Canvas
 - Allowed time: 2 hour
 - Connects with Kaggle competition
- Code portion
 - Also available Tuesday at noon in Canvas
 - Due end of day Sunday
 - Will include a test data set that you will submit prediction for

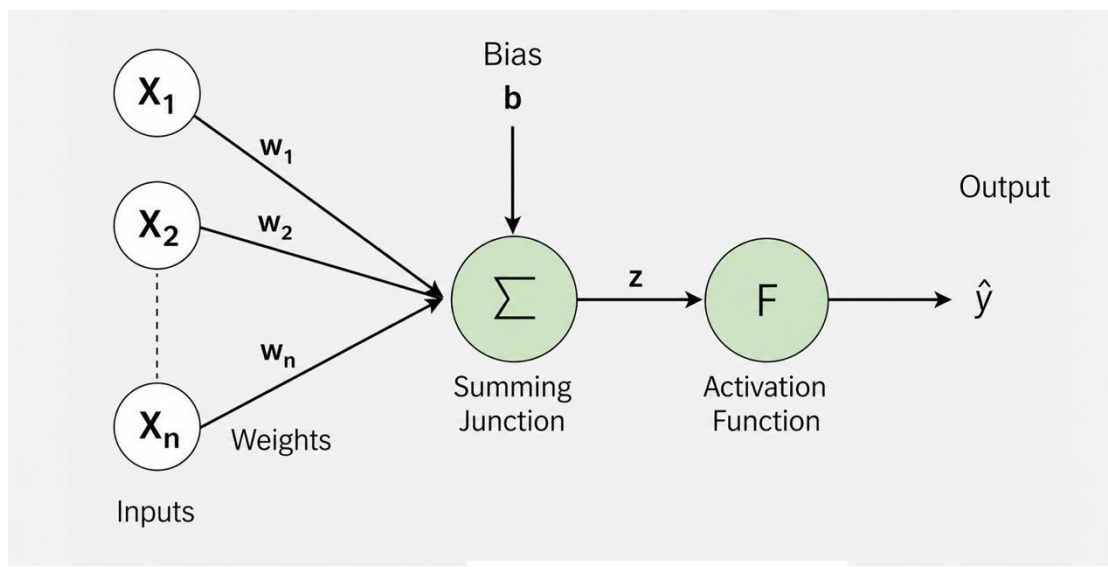
Neural network

$$y = w_1x_1 + w_2x_2 + b$$

Neural network



Neural network



$$z = w^T x + b$$

$$\hat{y} = f(z)$$

$$z = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

$$a = f(z)$$

Activation functions

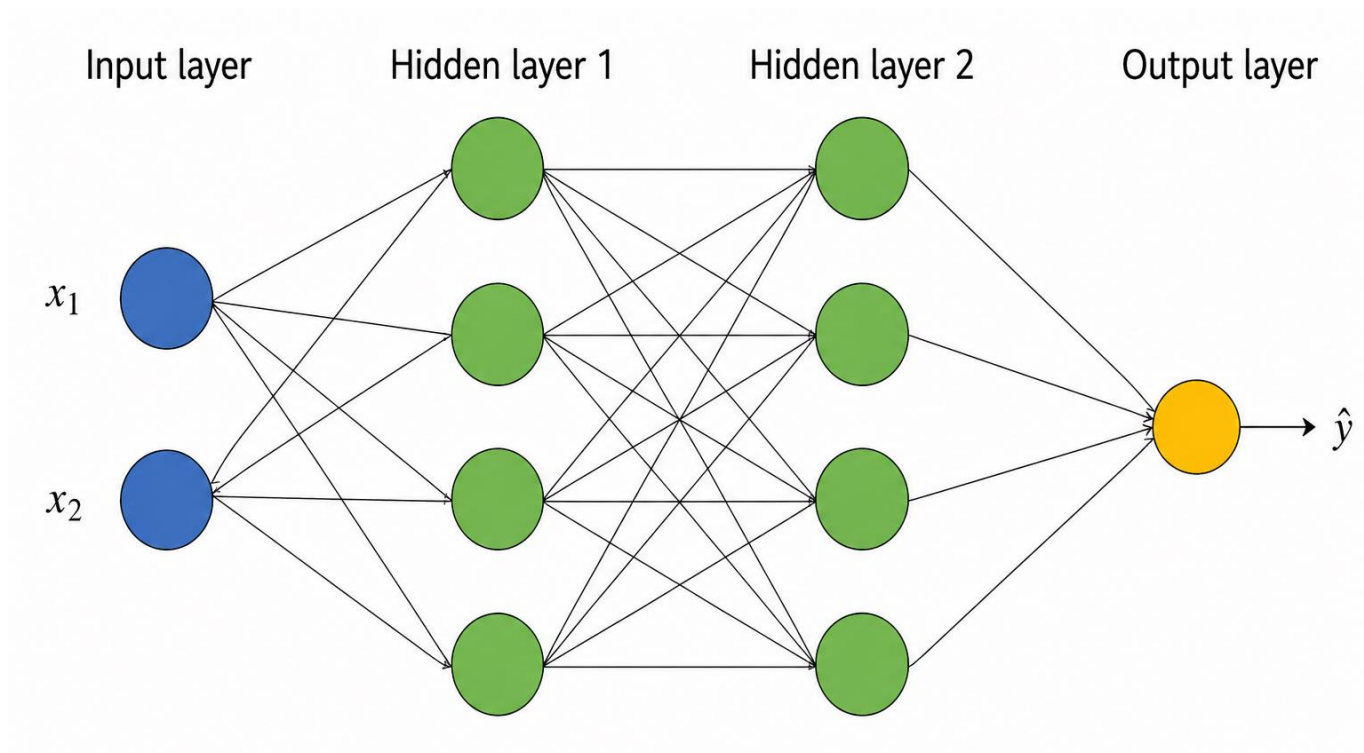
- Transforms linear output

$$a = f(z)$$

- ReLU (rectified linear unit)

$$f(z) = \max(0, z)$$

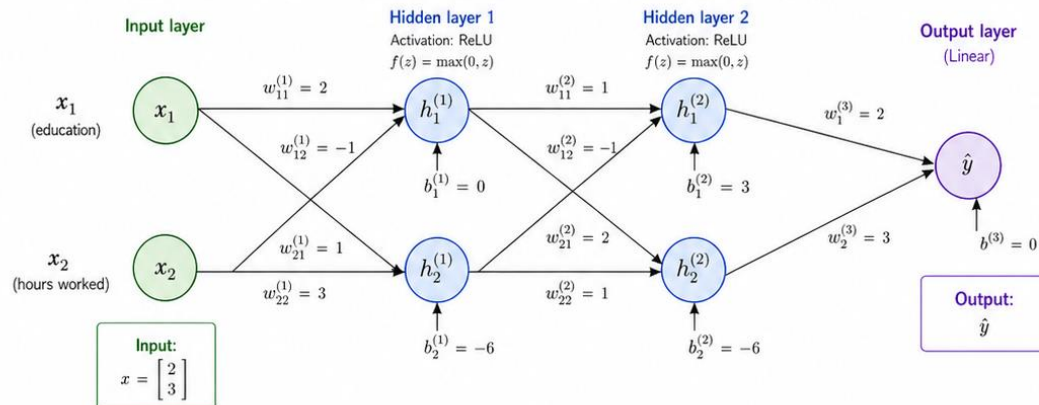
Multilayer perceptron = MLP



Running a neural network

WORKSHEET: Running a Neural Network by Hand

Use the network diagram below and the given weights, biases, and activation function to compute the output $\hat{y}$ for the input x .



PART 1: Forward pass (ReLU)

1. Compute the pre-activation values for Hidden layer 1:

$$z^{(1)} = W^{(1)}x + b^{(1)}$$

Neuron	Calculation	$z^{(1)}$
$h_1^{(1)}$		
$h_2^{(1)}$		

2. Apply ReLU to get the activations of Hidden layer 1:

$$a^{(1)} = f(z^{(1)})$$

Neuron	$z^{(1)}$	$\text{ReLU}(z^{(1)}) = a^{(1)}$
$h_1^{(1)}$		
$h_2^{(1)}$		

3. Compute the pre-activation values for Hidden layer 2:

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}$$

Neuron	Calculation	$z^{(2)}$
$h_1^{(2)}$		
$h_2^{(2)}$		

4. Apply ReLU to get the activations of Hidden layer 2:

$$a^{(2)} = f(z^{(2)})$$

Neuron	$z^{(2)}$	$\text{ReLU}(z^{(2)}) = a^{(2)}$
$h_1^{(2)}$		
$h_2^{(2)}$		

5. Compute the output (linear):

$$\hat{y} = W^{(3)}a^{(2)} + b^{(3)}$$

Calculation	$\hat{y}$ (final output)

PART 2: Interpretation

1. Which neurons in Hidden layer 1 are active (nonzero)? Which are "dead" (zero)?
2. Which neurons in Hidden layer 2 are active? Which are "dead"?
3. What information might be lost when a neuron outputs 0 due to ReLU?

PART 3: Extension – Leaky ReLU

Use Leaky ReLU:

$$f(z) = \begin{cases} z, & \text{if } z > 0 \\ 0.1z, & \text{if } z \leq 0 \end{cases}$$

Repeat the forward pass with Leaky ReLU in the hidden layers. What is the new output $\hat{y}$?

New $\hat{y}$ with Leaky ReLU:

PART 4 (Challenge): Identity activation

Suppose instead of ReLU we use the identity function $f(z) = z$ in both hidden layers. Compute the output $\hat{y}$. What kind of model does this network become?

$\hat{y}$ with identity activation:

What kind of model?

Leaky ReLU function

$$f(z) = \begin{cases} z & \text{if } z > 0 \\ \alpha z & \text{if } z \leq 0 \end{cases}$$

One epoch training cycle

- Forward Pass
- Loss
- Backpropagation
- Update
- Repeat

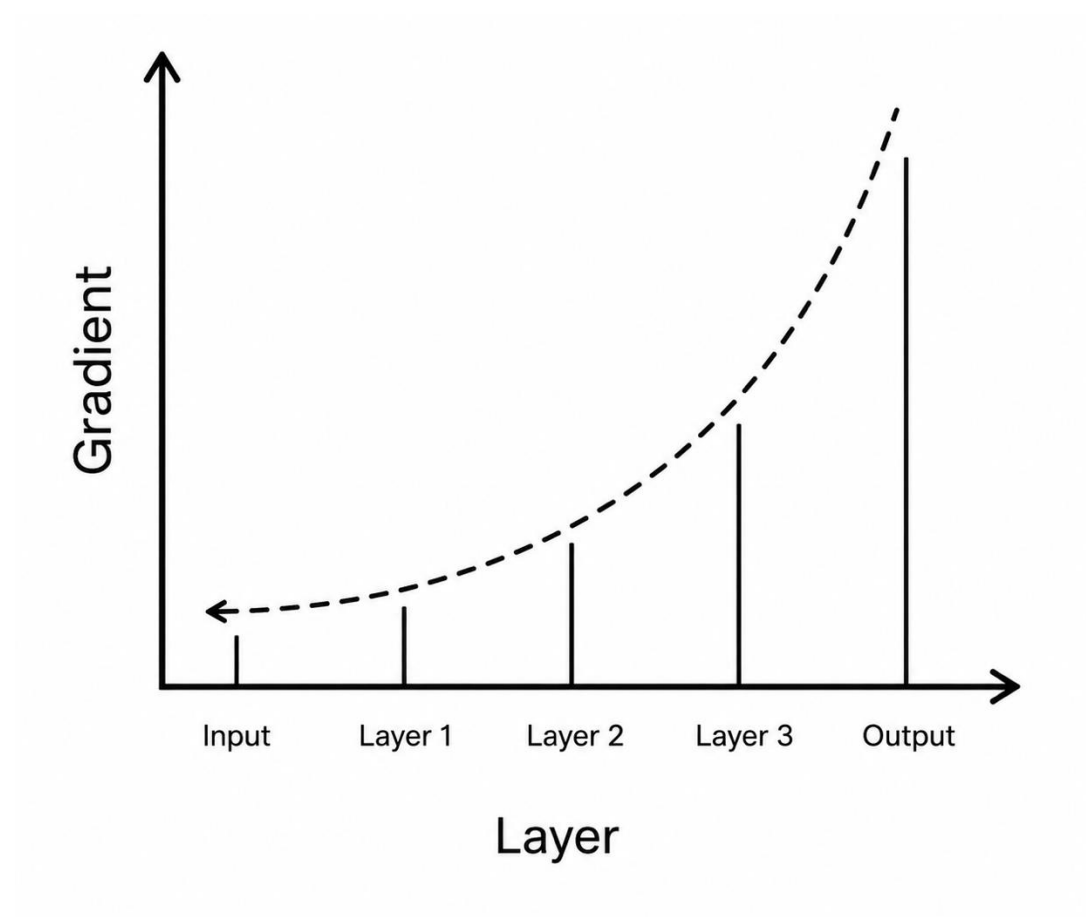
Backpropagation

$$L(y, \hat{y}) = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

$$L(y, \hat{y}) \quad \text{where} \quad \hat{y} = f(wx + b)$$

$$\frac{\partial L}{\partial w}$$

The vanishing gradient problem



Activation functions

- Sigmoid $\sigma(z) = \frac{1}{1 + e^{-z}}$
- tanh $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$
- ReLU $f(z) = \max(0, z)$
- Leaky ReLU $f(z) = \begin{cases} z & \text{if } z > 0 \\ \alpha z & \text{if } z \leq 0 \end{cases}$
- Softmax $\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$

Some tunable parameters

- Epochs
- Mini-batch size
- Learning rate
- Activation function

Batch size & gradient descent

- Stochastic gradient descent
- Mini-batch gradient descent
- Batch gradient descent

How many epochs?

- Small/simple problems
- Medium models
- Deep learning models